

AeroFlux Stage 2: A Scalable, Portable Big-Data Platform for Real-Time Flight-Delay Intelligence

Final Project Report

Jonathan Wilson

Ryan Sollom

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Instructor: Dr. Duoduo Liao

Institution: George Mason University

Abstract

TODO: abstract goes here (write last).

1 Introduction

Flight delays are among the most costly and disruptive failures in the U.S. National Airspace System (NAS), and they rarely occur in isolation. A single late arrival propagates downstream through the same aircraft’s rotation and through shared airport departure and arrival queues, so a local disturbance becomes a network-wide disruption.

Although airlines, airports, and air traffic authorities increasingly use real-time and predictive decision-support systems, these capabilities are often specialized around particular organizations, operational domains, or stages of flight. For example, the FAA’s Traffic Flow Management System supports demand-and-capacity management across the National Airspace System, while Terminal Flight Data Manager integrates surface and national traffic-management data to improve airport operations ([Federal Aviation Administration, 2020, 2024c](#)). Commercial platforms such as Sabre Mosaic support airline disruption recovery, and Assaia ApronAI uses computer vision and machine learning to monitor aircraft turnarounds and predict departure readiness ([Assaia International AG, n.d.](#); [Sabre Corporation, n.d.](#)). Opportunities therefore remain for integrated research platforms that fuse multiple live aviation data streams, model delay propagation across the broader flight network, and support forecasting, simulation, data science, and AI-assisted analysis.

1.1 Problem Statement

1.1.1 The Problem: Cascading Failure in Tightly-Coupled Systems

A **flight delay** — an arrival later than scheduled, conventionally by 15 or more minutes — is rarely an isolated event. Aircraft fly sequential legs, so a late inbound flight delays its own next departure (**rotational propagation**); airports have finite runway and gate capacity, so congestion at one flight’s origin ripples to every flight sharing that resource (**queue propagation**). Through these mechanisms, a single local disturbance becomes a network-wide disruption. This is **delay propagation**, and it matters operationally because the window to act on a disruption closes as it spreads — by the time a cascade is visible in aggregate, the cheapest interventions are already gone.

Compounding this, live operational data arrives as **heterogeneous, imperfect streams** — FAA SWIM flight messages, ADS-B aircraft positions, weather observations — that share no universal key and are measured differently from the historical records used to train models (**distribution shift**). Fusing these into a single, queryable operational view requires backward-looking temporal joins, UTC normalization, spatial matching, and entity resolution across systems that don’t agree on identity. This is the same class of real-time, multi-sensor data-fusion problem now central to edge AI, autonomy, and robotics: large, fast, noisy signals that must be fused and reasoned over in real time.

1.1.2 Contribution

Complex systems fail when disruptions propagate faster than their defenses can adapt. Reason’s Swiss Cheese model frames accidents as the momentary alignment of holes across successive layers of defense — but the classic model is static. It names the holes; it does not show which are aligning, or how fast. In a live, tightly-coupled system, the holes are *moving*.

AeroFlux’s contribution is a **dynamic defensive layer**: one that fuses live signals to see the holes forming, predicts their alignment, and gives operators time to act. Concretely, this means (1) real-time fusion of heterogeneous aviation and weather streams into a canonical, propagation-aware per-flight state; (2) predictive modeling of emerging delay risk with the network context (rotation, demand, weather) that drives propagation; and (3) an explainable analyst layer that makes those predictions inspectable in natural language.

We use the U.S. National Airspace System as a **proving ground**. Flight delay was chosen deliberately — not because it is unsolved, but because it is well-studied, has abundant public ground truth, and keeps prediction errors low-stakes, giving rapid feedback for developing and validating methods. The delay problem is the instance; the contribution is the *approach* — a real-time, network-aware intelligence layer for anticipating cascading consequences in tightly-coupled systems — which is designed to extend to other safety-critical cyber-physical domains.

1.1.3 Scope and Vision

This stage focused primarily on building the **infrastructure** such a layer requires: portable real-time ingestion and fusion, a train/serve-consistent feature pipeline, cloud-backed serving, live evaluation, and an explainable agent. It is foundation, not final form.

The longer-term vision is cyber-physical situational awareness for complex adaptive systems — emphasizing data fusion, propagation modeling, downstream- effect reasoning, and decision intelligence. Natural next directions include network-aware simulation, an operational digital twin of the airspace, and agentic reasoning over fused state; further out, the same fusion-and-propagation core could extend beyond aviation toward a multi-domain autonomy and decision-support layer. We hold these directions as directions, not commitments — broad, foundational work of this kind is most valuable when it leaves room for the solution space to reveal where it should go next.

Positioning. AeroFlux is explicitly *not* a replacement for certified collision-avoidance (ACAS/ATAS), air traffic control, or pilot tools. It is a higher-level, pre-tactical **fusion, simulation, and decision-support layer** that complements them — turning operations from reactive to anticipatory rather than intervening in the tactical safety loop.

1.2 Technical Challenges

Building this platform requires solving several non-trivial problems: (1) ingesting heterogeneous, error-prone streaming sources (FAA SWIM XML, ADS-B, weather) without data loss; (2) *entity resolution* across systems that share no universal key — for example, the

aircraft tail number is not published in SWIM flight data and must be recovered from ADS-B, which itself has limited coverage; (3) unstable identifiers, since a flight’s `flight_ref` changes on plan amendment while its Globally Unique Flight Identifier (GUFI) remains stable; (4) train/serve skew, because historical labels (Bureau of Transportation Statistics) and live features are measured differently; and (5) delivering all of this at demo quality on a portable, low-cost footprint.

2 Related Work

Machine-learning delay prediction. Prior work frames delay prediction as supervised classification or regression over historical records. The review by Sternberg et al. (2017) describes the major approaches used for flight-delay prediction, while more recent work continues to evaluate machine-learning algorithms and class-imbalance handling (AlBassam & AlShahrani, 2025). These approaches can achieve strong predictive performance on curated historical datasets. However, when used in isolation, they may remain largely retrospective, rely on a limited set of data sources, and model flights as independent observations. These assumptions can restrict their ability to capture the temporal, spatial, and operational dependencies through which delays propagate across the aviation network.

Delay-propagation modeling. A separate literature models how delays ripple through airline schedules. Beatty et al. (1999) introduced an early method for evaluating delay propagation through an airline schedule, Wong & Tsai (2012) applied a survival model, and Kaffle & Zou (2016) proposed an analytical-econometric approach. These methods capture important propagation mechanisms but are generally offline and disconnected from real-time streaming prediction. Belcastro et al. (2016) demonstrated scalable data mining for flight-delay prediction, establishing the value of big-data tooling while leaving room for tighter integration with real-time data fusion.

Operational data fusion. The state of the art in operational systems includes NASA’s Airspace Technology Demonstration 2 (ATD-2) *Fuser*, which fuses multiple aviation data sources into a unified flight record (National Aeronautics and Space Administration, 2021). Its strength is production-grade fused operational data; its limitation for this project is that it is not designed as a lightweight, open, extensible ML/AI experimentation platform. This project leverages some of the high-level concepts used in ATD-2 to create its own fuser.

Emerging aviation intelligence systems. Recent operational systems show that the field is moving beyond static delay prediction toward real-time prediction, simulation, and decision support. FlightAware Foresight represents production ML-driven predictive flight tracking, including arrival-time, taxi-out, and runway predictions (FlightAware, {n.d.}). NASA’s NAS Digital Twin and Project Bluebird extend this direction toward replayable and probabilistic simulation environments for live or historical airspace operations and AI-agent evaluation (Lauderdale et al., 2024; Pepper et al., 2026). Commercial platforms such as Airspace Intelligence PRESCIENCE, Lufthansa Systems NetLine/Ops++ aiOCC, and Amadeus Flight Operations Control show the market moving toward 4D operating-domain twins, AI-assisted operations control, disruption recovery, and resource-aware decision support (Airspace Intelligence, {n.d.}; Amadeus, {n.d.}; Lufthansa Systems, 2025). Recent

prototypes such as FlightSense further combine rotation-chain features, real-time MLOps, and conversational/agent interfaces for delay prediction (Shelke et al., 2026), while the FAA AI safety roadmap highlights the need for assurance, incremental deployment, and aviation-specific safety methods when adding AI to operational contexts (Federal Aviation Administration, 2024a). Together, these efforts motivate AeroFlux as a lightweight, SWIM-native experimentation platform that connects real-time data fusion, propagation-aware ML, replay, and explainable/agent analysis rather than treating delay prediction as a standalone model.

Relevance of this approach. AeroFlux Stage 2 bridges these strands. It couples lightweight, Fuser-inspired real-time fusion and entity resolution with propagation-aware feature engineering and an explainable retrieval-augmented agent — integrating capabilities that prior work often treats separately.

3 Data and Fusion

3.1 Data Sources

The platform fuses live and historical aviation and weather sources, plus reference dimensions and a documentation corpus, summarized in Table 1. FAA SWIM provides the authoritative live NAS flight state (Federal Aviation Administration, 2024b); ADS-B supplies the physical airframe identity (ICAO 24-bit address and registration) needed to link an aircraft’s successive legs, drawn from community feeds (adsb.lol, 2026; airplanes.live, 2026), whose research value is demonstrated by large-scale networks such as OpenSky (Schäfer et al., 2014); NOAA weather — a leading delay driver — is drawn from live METAR via the Aviation Weather Center (National Oceanic and Atmospheric Administration, 2026) for serving and the NCEI Integrated Surface Database for historical training (National Centers for Environmental Information, 2026); and the BTS On-Time Performance dataset, which includes actual gate times and tail numbers, provides historical training labels (Bureau of Transportation Statistics, 2026). Reference dimensions for airports and airlines (OpenFlights, 2026; OurAirports, 2026) support entity resolution, code normalization, and timezone alignment.

Table 1: AeroFlux Stage 2 data sources, retention policies, and estimated storage requirements. Estimated sizes are project-planning ranges rather than published source totals and depend on geographic scope, ingestion frequency, selected fields, compression, and duplicate handling.

Source	Type	Role	Retention	Estimated retained size
FAA SWIM (TFMS)	Streaming XML	Live flight state (primary)	48 h	Approximately 5–25 GB
ADS-B (airplanes.live / adsb.lol)	Streaming JSON	Airframe and tail resolution	48 h	Approximately 2–10 GB

Source	Type	Role	Retention	Estimated retained size
NOAA weather — METAR (AWC) + NCEI ISD hourly	Polled JSON / text	Weather: METAR for serving, NCEI for training	48 h	Approximately 50–250 MB live
BTS On-Time Performance	Historical CSV / Parquet	Historical training labels	Persistent	Approximately 20–40 GB as CSV or 4–10 GB as Parquet
Aviation documents (FAA, SWIM dictionaries, and related references)	PDF / HTML / text	RAG documentation corpus	Persistent	Approximately 0.5–2 GB, including extracted text and embeddings
Reference dimensions (airports, airlines)	Static CSV	Entity resolution, ICAO/IATA + timezone normalization	Persistent	< 50 MB

3.2 Fusion and Canonical Records

A raw SWIM message (abridged) is shown in Listing 1; the corresponding fused, validated canonical record produced by the ingestion pipeline is shown in Listing 2. Each SWIM document explodes into one record per flight, which the pipeline fuses by GUF1 across many messages over time.

Listing 1 Raw SWIM TFMS flight message (abridged).

```
<fdm:fltMessage acid="AAL2033" airline="AAL" arrArpt="KLGA" depArpt="KCLT"
  flightRef="150976233" msgType="FlightModify" ...>
  <nxc:flightStatus>PLANNED</nxc:flightStatus>
  <nxc:aircraftModel>A320</nxc:aircraftModel>
  <nxc:flightTimeData airlineOutTime="2026-07-04T15:39:00Z"
    airlineInTime="2026-07-04T17:35:00Z"/>
</fdm:fltMessage>
```

TODO: expanded fusion description and tier diagram/figure.

Listing 2 Fused, schema-validated canonical flight-instance record.

```
{ "flight_instance_id": "KN6770564K", "callsign": "AAL2033",  
  "flight_number": "AA2033", "carrier_name": "American Airlines",  
  "origin": "KCLT", "destination": "KLGA", "aircraft_type": "A320",  
  "scheduled_gate_departure": "2026-07-04T15:39:00Z",  
  "estimated_arrival": "2026-07-04T17:41:00Z", "flight_status": "PLANNED",  
  "resolution_status": "airline_resolved", "schema_version": "1.0" }
```

4 Methods

4.1 System Architecture

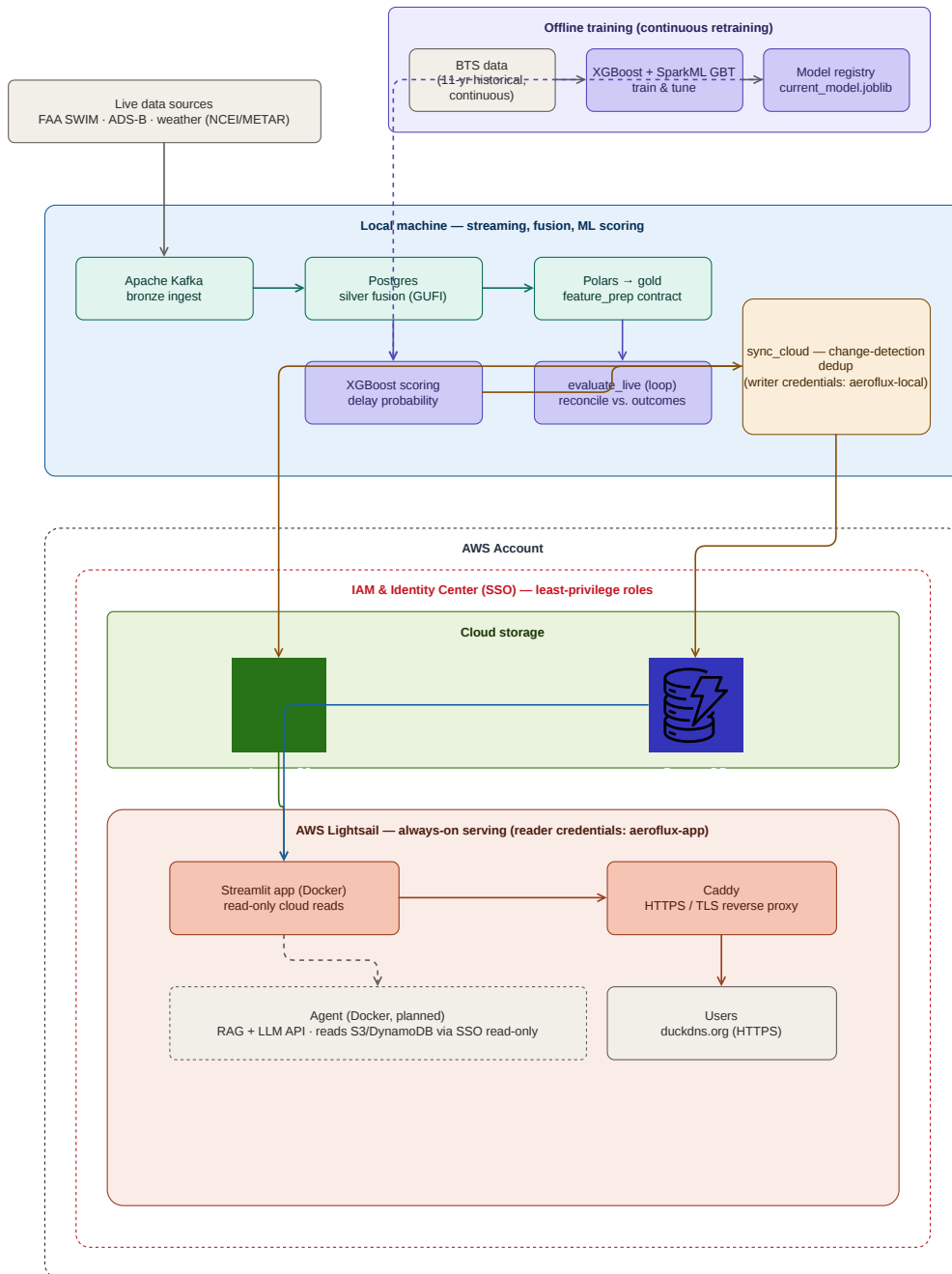


Figure 1: AeroFlux Stage 2 system architecture.

TODO: architecture prose.

4.2 Ingestion and Streaming

TODO: ingestion + streaming.

4.3 Feature Engineering

TODO: feature engineering + the parity argument.

4.4 Modeling

TODO: models.

4.5 Training and Validation Strategy

TODO: train/test strategy, tuning, leakage prevention.

4.6 Live-Prediction Evaluation

TODO: live-eval methodology.

4.7 Agent Prototype

TODO: agent prototype (coordinate with Ryan).

5 Evaluation and Results

We evaluate at two levels, each against an explicit baseline: predictive (model) performance and system performance.

5.1 Predictive Performance

TODO: predictive results — BTS vs. live comparison table + calibration curve.

Table 2: Predictive performance, historical training vs. live reconciled evaluation.

Metric	BTS training (held-out)	Live (reconciled)
ROC-AUC	<i>TODO</i>	<i>TODO</i>
PR-AUC	<i>TODO</i>	<i>TODO</i>
F1	<i>TODO</i>	<i>TODO</i>
Brier	<i>TODO</i>	<i>TODO</i>
n	<i>TODO</i>	<i>TODO</i>

5.2 Feature Importance

TODO: feature importance figure + discussion.

5.3 System Performance

TODO: system performance table (baseline vs. achieved).

5.4 Expected vs. Actual Performance Targets

TODO: targets vs. achieved table.

6 Discussion

6.1 Key Concepts and Definitions

TODO: concepts + glossary.

6.2 Assumptions

TODO: assumptions.

6.3 Limitations

TODO: limitations (distribution shift, censoring, coverage, parity gaps).

6.4 What Was Proposed vs. Delivered

TODO: proposed vs. delivered table.

7 Technology Stack

Table 3: Technology stack and how each component was used.

Category	Tool	How it was used
Language	Python 3.11	<i>TODO</i>
Streaming	Apache Kafka	<i>TODO</i>
Processing	Polars (Spark Structured Streaming as scale path)	<i>TODO</i>
Storage (local)	PostgreSQL	<i>TODO</i>
Storage (cloud)	Amazon S3, DynamoDB	<i>TODO</i>
ML	scikit-learn, XGBoost, SparkML GBT, SHAP	<i>TODO</i>
Serving / UI	Streamlit, Plotly	<i>TODO</i>

Category	Tool	How it was used
Deployment	Docker, Caddy, GHCR, GitHub Actions, AWS Lightsail	<i>TODO</i>
Security	AWS IAM, Identity Center (SSO)	<i>TODO</i>
AI orchestration	LangGraph, RAG, LLM API (Bedrock/Anthropic)	<i>TODO (Ryan)</i>

8 Conclusion and Future Work

TODO: conclusion + future work.

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